

Chris Moore

Maker & Engineer — Hardware, Firmware, Full-Stack Software & Manufacturing

Henderson, NV / (702) 752-9213 / chrismoore044@gmail.com / chris-moore.dev / github.com/Grudged

I design and build complete products end-to-end — custom C firmware and wireless protocols, 3D-printed enclosures modeled in Fusion 360, and the polished software that wraps around them. I also bring something most engineers don't: I've **worked the manufacturing floor**, helping run two production lines and their large assembly teams building complex electromechanical devices at scale — so I understand what it takes to carry a product from prototype to volume. Off the clock I'm always building something: wireless hardware, 3D-printed gadgets, and full-stack web apps that turn complicated things into tools people actually enjoy using.

FEATURED BUILD — HARDWARE

Wireless Co-Op Handheld — personal hardware build, write-up at chris-moore.dev

- Two-player handheld gaming system designed from scratch — each player gets their own wireless screen and controller streaming from one console; a Wii U GamePad concept rebuilt from ~\$15 parts.
- Wrote the controller firmware in C (Pico 2W / pico-sdk): oversampled analog sticks, debounced buttons, boot-time auto-calibration, packed into a 20-byte UDP protocol at 60 Hz with zero packet loss room-to-room. Receiver injects a virtual gamepad via Linux uinput.
- Engineered a documented H.264 latency budget (~15–30 ms, on par with the original silicon), a full bill of materials, and a 3D-printed enclosure & mounting bracket through several printed revisions.

EXPERIENCE

Light & Wonder (formerly Scientific Games / Bally Technologies)

11 years across engineering, data, and manufacturing · Oct 2014 — Present

Software Engineer

Apr 2023 — Present

- Build and maintain scalable data pipelines and backend processes for real-time enterprise application integration; optimized PL/SQL procedures to cut system latency by 30%+.
- Build internal dashboards (Angular, JavaScript) and ETL workflows (SQL, Python) that turn raw operational data into tools teams actually use.
- Work with DevOps, QA, and Product to translate needs into maintainable software, and lead integration testing through deployment.

Sr. Business Analyst / Data Engineer

Jul 2018 — Apr 2023

- Designed and managed data pipelines supporting analytics and decisions across departments; delivered reporting on inventory trends and logistics performance.
- Documented key operational processes and drove workflow improvements across enterprise systems.

Floor Supervisor — Electromechanical Product Assembly ★

Feb 2016 — Jul 2018

First-hand experience in what it takes to build a complex physical product at volume.

- Co-supervised **two production lines** and their large assembly teams building **card shufflers and electronic gaming machines** — precision electromechanical devices integrating mechanics, electronics, and embedded displays.
- Scheduled resources, owned quality control, and consistently hit production milestones and budgets.
- Improved throughput through strategic staffing, workflow redesign, and Kaizen / continuous-improvement practices — direct fluency in design-for-manufacturing and assembly.

Technical Analyst

Oct 2014 — Feb 2016

- Resolved user-reported system issues and found long-term fixes for recurring problems; supported new system rollouts with documentation, testing, and continuous-improvement feedback.

SELECTED PROJECTS

StarBrain Academy — original hands-on learning app for my 5-year-old; project-based, low-pressure science play. [Web](#) / [AI](#)

FirstEmbark — free, no-login career platform matching new grads to real careers and nearby training. [FastAPI](#) · [O*NET](#)

Open Scholarships — free, ad-free student-aid directory published as open, machine-readable data. [FastAPI](#) · [CC BY 4.0](#)

Reef Watch — interactive 3D globe visualizing coral-reef health from real satellite imagery. [CesiumJS](#)

CharacterQuill — creative companion for fiction writers: character building, in-voice AI feedback, visual plot maps. [Angular](#) · [AI](#)

Mission Control — a 23-view PWA running all my infrastructure; making complex systems legible. [Angular](#) · [FastAPI](#)

TOOLBOX

Hardware & making: C firmware (Pico/RP2040, ESP32, Arduino), Raspberry Pi, soldering & breadboarding, sensors & ADC, Fusion 360 CAD, 3D printing (STL), UDP/BLE protocols, latency & power budgeting.

Manufacturing: assembly-line supervision, QC, Kaizen / continuous improvement, design-for-manufacturing, production scheduling, Oracle WMS, MAPICS ERP, PLM.

Software: Python (FastAPI), Angular/TypeScript, JavaScript, SQL & PL/SQL (Oracle, Postgres, SQLite), Power Platform, REST APIs, Git / CI-CD pipelines, technical writing & build documentation.

EDUCATION & CERTIFICATIONS

A.A.S., Computer & Information Technology — Software Programming — College of Southern Nevada, in progress (33 credits)

Microsoft Office Specialist – Associate (2025)

Management Certification · Kaizen Training

The Complete JavaScript Course – Udemy (2025)

Angular – The Complete Guide – Udemy (2024)

The Complete Oracle SQL Bootcamp – Udemy (2025)

Understanding TypeScript – Udemy (2024)